S1 Text.

ODD description of EvoSim

Supporting information for "Environmental fluctuation erodes gene regulation unless it can be tracked"

This description follows the ODD protocol in its 2020 revision (Grimm et al., J Artif Soc Soc Simul 23(2):7). It documents the version of EvoSim used for the experiments in the main text, and every parameter value is given in Table 1. Where a mechanism is switched off in these experiments it is described briefly and marked as unused.

1. Purpose and patterns

EvoSim is an individual-based model of unicellular organisms whose genomes are gene regulatory networks of evolvable size and connectivity. In this study it is used to ask how environmental fluctuation, and the information available to track it, affect the number of regulatory connections and their strength. The model is built so that network size and connectivity evolve under mutation, selection and drift. Each phenotypic trait has a cost as well as a benefit, so traits settle at interior values, and the resource ecology is built partly by the organisms themselves through the byproducts of consumption.

2. Entities, state variables and scales

Grid cells. The environment is a square lattice of $G \times G$ cells ($G = 128$ in this study) with reflecting boundaries. Each cell holds the concentration of six resource channels, $R_r(x, y)$ for $r = 1, \ldots, 6$, capped at 6.0, and a count of the organisms occupying it (at most 8).

Organisms. Each organism has a position, an energy store (at most 4.0), an age in time steps, a lineage depth (number of ancestral divisions), a deleterious-load counter, and a genome. The genome is a regulatory network with n_g nodes, $13 \leq n_g \leq 32$, consisting of a weight matrix $W \in \mathbb{R}^{n_g \times n_g}$ and a bias vector $b \in \mathbb{R}^{n_g}$. Nodes 1–12 are the phenotype outputs, node 13 is the sensor for the abiotic condition, and nodes $14, \ldots, n_g$ are hidden nodes with no fixed meaning. Each organism also carries its phenotype, computed once at birth and fixed for life in these experiments. It consists of twelve traits in $[0, 1]$, namely six resource-uptake shares (normalised to sum to one), body size, motility, division threshold, DNA-repair fidelity, stress tolerance and leakage.

Global state. The abiotic condition, which cycles through discrete states (Section 7.3); the seasonal phase; and a famine counter.

Scales. Time advances in discrete steps. In the experiments of this study one generation took about 24 time steps in fluctuating and 31 in frozen populations (measured from the slope of mean lineage depth over the second half of each run), so a 150,000-step run covers roughly 5,000 to 6,000 generations. The maximum age is 60 steps.

3. Process overview and scheduling

Each time step executes the following steps in order.

1. *Resource renewal.* Every channel in every cell decays, and external inflow is added (Section 7.1).

2. *Shock check.* With a small probability a mass-mortality or famine event occurs (Section 7.9).

3. *Reversible development* (Section 7.4), which was not used in this study.

4. *Organism update.* Organisms are visited once each, in storage order. For each organism the step comprises uptake and conversion of resources, with byproduct release into the cell (Section 7.5); maintenance cost (7.6); energy update; ageing unless dormant (7.7); movement (7.8); death if energy is exhausted or the maximum age is exceeded; otherwise, if the energy store exceeds the division threshold and the organism is not dormant, it is queued to divide.

5. *Births* (Section 7.10). Queued parents are processed in queue order. Placement is serial; mutation and development of the offspring are computed in parallel with a random seed drawn serially for each offspring, so results do not depend on the number of threads.

6. *Diffusion* of all channels (Section 7.2).

Because organisms are updated sequentially and draw on shared cell resources, an organism visited earlier in a step can deplete a cell before a later one reaches it. Storage order reflects the reuse of memory slots and has no biological meaning.

4. Design concepts

Basic principles. Every trait that brings a benefit also carries a cost, so that selection does not drive traits to a boundary. Uptake is an allocation of a fixed budget across the six channels, so the affinities are not independent. Consumption of channel r releases a byproduct into channel $r + 1$, so all channels except the first exist only through the activity of the organisms.

Emergence. Network size, connectivity and connection strength, population size, resource distribution and the composition of the community emerge from the individual processes.

Adaptation. Organisms do not adapt within their lifetime. The phenotype is fixed at birth and is a function of the genome, the local resource concentrations and the value of the abiotic cue at the place and time of birth.

Objectives, learning and prediction. None; there is no explicit fitness function. Reproductive success follows from energy balance.

Sensing. At birth, an organism senses the concentrations of the six channels in its cell and the value of the abiotic cue.

Interaction. Organisms interact through shared resources, through byproducts that others can consume, through crowding costs among organisms in the same cell, and through the occupancy limit that can block a division.

Stochasticity. Mutation, movement, the placement of offspring, shocks and the founding genomes are random. All randomness derives from the simulation seed, and runs are exactly reproducible for a given seed.

Collectives. None are modelled explicitly. Lineages are tracked as clades for bookkeeping only; they do not affect the dynamics.

Observation. Population means of the phenotype, of network structure (number of nodes, number of non-zero weights, mean absolute weight) and of ecological quantities are recorded at every time step. The measures used in the main text are defined in Section 7.12.

5. Initialisation

All cells start with $R_1 = 1.0$ and the other channels empty. $\max(200, G^2/80)$ founders (204 at $G = 128$) are placed in uniformly random cells with energy 0.8, age 0 and load 0. Each founder has $n_g = 16$ nodes; each weight is non-zero with probability 0.25 and then drawn from $\mathcal{N}(0, 0.6^2)$, and biases are drawn from $\mathcal{N}(0, 0.2^2)$. Founders are developed in the conditions of their cell before the first step.

6. Input data

None. The model uses no external data.

7. Submodels

7.1 Resource renewal

Each step, every channel in every cell is multiplied by $1 - 0.006$. External inflow per cell is

$$I(t) = 0.25 \left(1 + 0.6 \sin \frac{2\pi t}{15{,}000}\right) \times f(t),$$

where $f(t) = 0.65$ during a famine and 1 otherwise, and $I(t)$ is set to zero if negative. In frozen environments (one condition state) all inflow enters channel 1. In fluctuating environments 60% enters channel 1 and 40% enters the channel with the highest yield in the current state (Section 7.3). Concentrations are capped at 6.0 after each addition.

7.2 Diffusion

Each channel diffuses by a five-point Laplacian with rate 0.12 per step and reflecting (no-flux) boundaries; concentrations are capped at 6.0.

7.3 Abiotic condition, state and cue

The condition has n_s states ($n_s = 6$ in fluctuating arms, 1 in frozen arms) and a period of $P = 2{,}000$ steps. With phase $\theta(t) = (t \bmod P)/P$, the current state is $s = 1 + \lfloor n_s \theta \rfloor$. The state rotates the yields, so that the yield of channel r in state s is $y_{1+((r-1+s-1) \bmod 6)}$, with $y = (1.00, 0.60, 0.45, 0.35, 0.28, 0.22)$, so a different channel pays best in each state. The cue delivered to node 13 is defined as follows.

- In the informative arm $c = 2\theta - 1$, which determines the state exactly.
- In the yoked arm $c = 2\phi - 1$, with ϕ the fractional part of $(t \bmod P) \cdot 6.1803399/P$. The multiplier $6.1803399 = 10/\varphi$, with φ the golden ratio, is irrational, so the cue covers the same range without locking to the cycle and carries 0.0004 bits of information about the state.
- In the no-cue arm $c = 0$.

With $n_s = 1$ the state is always 1 and the cue is zero, so the three frozen arms are identical.

7.4 Development

At birth the expression vector is initialised to $e_k = 0.1$ and updated synchronously eight times,

$$e_k \leftarrow \tanh\Big(b_k + \sum_{m=1}^{n_g} W_{mk}\, e_m + u_k\Big),$$

where $u_k = R_k$ at the organism's cell for $k \le 6$, $u_{13} = c$, and $u_k = 0$ otherwise. The phenotype is $\phi_k = (e_k + 1)/2$ for $k = 1, \ldots, 12$, and the uptake shares $\phi_1, \ldots, \phi_6$ are rescaled to sum to one. Development does not reach a fixed point in general; the phenotype is the state after the eighth update, which is deterministic and heritable. Reversible development, in which living organisms are re-developed every few steps, exists in the model but was not used in this study.

7.5 Uptake and conversion

Let $\ell = 0.45 + 0.55\,\phi_{12}$ be the effective leakage and $m = (0.5 + \phi_7)(1 + 1.2\,\ell)$ the rate multiplier (multiplied by 0.35 when dormant). For each channel r with share $a_r = \phi_r > 0$ and local concentration R_r, the amount taken is $g_r = \min(0.6\,R_r\,a_r\,m,\ R_r)$. It is removed from the cell, the organism gains $g_r\,y'_r\,(1 - \ell)$ energy, where y'_r is the current yield of channel r, and $g_r\,\ell$ is added to channel $r + 1$ in the same cell (the byproduct of channel 6 leaves the system). The summed concentration of channels 2–6 in the cell is recorded as the local toxicity T.

7.6 Maintenance

The energy cost per step is

$$0.08 + 0.22\,\phi_7^2 + 0.035\,\phi_8 + 0.04\,\phi_{11}^2 + 0.02\,\phi_{10} + 0.0015\,(n_g - 13) + \kappa_e\,n_{nz} \\ + 0.05\,(\text{other organisms in the cell}) + 0.015\,L + 0.008\,T\,(1 - \phi_{11}),$$

where n_{nz} is the number of non-zero weights, $\kappa_e = 5 \times 10^{-5}$ in these experiments, and L is the load counter. The cost is multiplied by 0.05 when dormant. Energy is updated as gains minus cost, capped at 4.0.

7.7 Dormancy and ageing

An organism whose energy at the start of the step is below 0.15 is dormant for that step. A dormant organism has reduced uptake (Section 7.5) and a reduced maintenance cost (Section 7.6), does not age and cannot divide. Otherwise its age increases by one.

7.8 Movement

With probability ϕ_8 the organism moves to a cell chosen uniformly from its Moore neighbourhood including its own cell, clamped at the boundary.

7.9 Shocks

At each step, with probability 1/25,000, one of three events occurs, each with probability 1/3: a cull, in which every organism dies with probability 0.92; a heat shock, in which each organism dies with probability $0.92\,(1 - \phi_{11})$; or a famine, which multiplies inflow by 0.65 for the next 400 steps. These events occur in all arms.

7.10 Division and inheritance

An organism that is not dormant and whose energy is at least $2.0\,(0.5 + \phi_9)$ divides. The offspring is placed in a random cell of the parent's Moore neighbourhood (including the parent's cell); if that cell already holds 8 organisms the division fails. Parent and offspring each receive half the parent's energy. The offspring has age 0, lineage depth one greater than the parent, and load equal to the parent's load plus the change from mutation (Section 7.11), floored at zero. Its genome is a mutated copy of the parent's, and it is developed in the conditions of its own cell.

7.11 Mutation

The per-site rate is $\mu = \max\left(10^{-3}(1 - 0.85\,\phi_{10}^{\text{parent}}),\, 10^{-6}\right)$, so DNA repair can reduce it by up to 85%. At each birth the offspring genome receives a Poisson number of point mutations with mean μn_g^2, each adding $\mathcal{N}(0, 0.25^2)$ to a randomly chosen weight, and with probability μn_g one randomly chosen bias receives $\mathcal{N}(0, 0.25^2)$ as well. With probability 0.2 (in these experiments; 0 by default) one randomly chosen weight is set to exactly zero. We call this connection silencing. Without it, additive mutation would drive every weight away from zero and density towards one.

Structural mutations act on whole nodes. With probability 0.002 all outgoing weights of a random node are set to zero (gene silencing). With probability 0.002 the incoming and outgoing weights and the bias of a random node are copied onto another random node, and $\mathcal{N}(0, 0.05^2)$ noise is added to every copied weight, including weights that were zero in the source (gene duplication). With probability 0.003 a hidden node is added, if $n_g < 32$, as a copy of a random existing node made in the same way as a duplication. If no node is added, a random hidden node is removed with probability 0.003, provided $n_g > 13$. In addition, each point mutation, bias mutation and connection silencing is classified for the load counter as deleterious (probability 0.70, load $+1$), beneficial (0.05, load -1) or neutral (otherwise), and each gene silencing adds 1 to the load. Load carries a maintenance cost (Section 7.6) and can fall as well as rise.

7.12 Measures used in the main text

Two quantities are computed for each organism at birth, the number n_{nz} of weights with $|w| > 10^{-4}$ and the strength density $\sum |w|/n_g^2$ over all n_g^2 weights, zeros included. At each time step the population means of n_{nz}, of strength density and of n_g are recorded. Count density is the mean n_{nz} divided by the squared mean n_g, and total strength $\sum |w|$ is the mean strength density times the squared mean n_g. Each run is summarised by the mean over its final 20% of time steps, with ratios computed per step before averaging.

Parameters

Table 1. EvoSim parameters. "Default" is the value in the code; "This study" lists values that differ from the default.

Parameter	Default	This study
Grid size G	256	128
Maximum organisms per cell	8	
Founders	$\max(200, G^2/80)$	204
Base inflow	0.25	
Season period, amplitude	15,000; 0.6	
Decay per step	0.006	
Diffusion rate	0.12	
Resource cap	6.0	

Parameter	Default	This study
Yields $y_1, \ldots, y_6$	1.00, 0.60, 0.45, 0.35, 0.28, 0.22	
Condition states n_s	1	6 (fluctuating), 1 (frozen)
Condition period P	2,000	
Inflow share following the favoured channel	0	0.4 (fluctuating)
Cue gain	1	0 (no-cue arms)
Yoked cue	off	on (yoked arms)
Uptake rate	0.60	
Leakage floor; rate bonus	0.45; 1.2	
Maintenance per step (base, size^2, motility, stress^2, repair)	0.08, 0.22, 0.035, 0.04, 0.02	
Maintenance per hidden node	0.0015	
Maintenance per non-zero weight κ_e	0	5×10^{-5}
Maintenance per step (crowding, load, toxicity)	0.05, 0.015, 0.008	
Division energy	2.0	
Initial energy; maximum energy	0.8; 4.0	
Dormancy threshold; upkeep; uptake	0.15; 0.05; 0.35	
Maximum age	60	
Development iterations	8	
Per-site mutation rate; σ	10^{-3}; 0.25	
Maximum repair suppression	0.85	
Gene silencing; duplication	0.002; 0.002	
Node addition; deletion	0.003; 0.003	
Connection silencing per birth	0	0.2
Deleterious; beneficial mutation share	0.70; 0.05	
Initial nodes	16	
Shock probability per step; kill fraction	1/25,000; 0.92	
Famine length; inflow factor	400; 0.65	
Run length		150,000 steps
Seeds		1–30 per arm

S2 Text.

R-Aevol build, configuration and the yoked-control derivation

Supporting information for "Environmental fluctuation erodes gene regulation unless it can be tracked"

1. Software and build

We used the regulation-enabled build of Aevol (R-Aevol) from the `aevol-7` branch of https://gitlab.inria.fr/aevol/aevol.git, commit `33277374`. The default branch (`aevol-9`) does not include regulation. Regulation is a compile-time option that defines `__REGUL`; its sources are in `src/libaevol/raevol/`. The build used CMake with

```
cmake .. -Dwith-raevol=ON -Dwithout-stats=OFF -Dwith-x=OFF \
         -DCMAKE_BUILD_TYPE=Release -DCMAKE_POLICY_VERSION_MINIMUM=3.5 \
         -DCMAKE_CXX_FLAGS="-Xpreprocessor -fopenmp -I/opt/homebrew/opt/libomp/include" \
         -DCMAKE_EXE_LINKER_FLAGS="-L/opt/homebrew/opt/libomp/lib -lomp"
```

`-Dwithout-stats=OFF` is required for the per-generation statistics file used in all analyses; the option is on by default and suppresses it. Building on Apple Silicon (macOS, Apple clang 21, Homebrew `cmake`, `libomp` and `boost`) needed changes that do not affect the model. The policy flag above is required because the bundled GoogleTest declares a CMake minimum below 3.5. The OpenMP include directory had to be passed through `CMAKE_CXX_FLAGS`, since it did not reach every target otherwise. Line 118 of `CMakeLists.txt` hard-codes the Intel Homebrew path `/usr/local/lib/libomp.dylib`, which we changed to `/opt/homebrew/opt/libomp/lib/libomp.dylib`. All simulations ran single-threaded, ten or twelve at a time, on one 12-core machine.

2. Parameter file

Each run was created with `aevol_create` from a parameter file generated by `replication/raevol/gen_params_v2.py`. The exploratory runs used `gen_params_v1.py` and the grain experiment `gen_params_grain.py`, which imports the same template. The template, with the fields that vary between runs shown in braces, is:

```
STRAIN_NAME             {name}
SEED                    {seed}
INIT_POP_SIZE           {1024 or 512}
WORLD_SIZE              32 {32 or 16}
INIT_METHOD             ONE_GOOD_GENE CLONE
CHROMOSOME_INITIAL_LENGTH   5000
FUZZY_FLAVOR            3
SELECTION_SCHEME        fitness_proportionate   750
POINT_MUTATION_RATE     5e-5
SMALL_INSERTION_RATE    5e-5
SMALL_DELETION_RATE     5e-5
MAX_INDEL_SIZE          6
WITH_ALIGNMENTS         false
DUPLICATION_RATE        5e-5
DELETION_RATE           5e-5
TRANSLOCATION_RATE      5e-5
INVERSION_RATE          5e-5
BACKUP_STEP             10000
RECORD_TREE             false
HILL_SHAPE_N                    4
HILL_SHAPE_THETA                0.5
DEGRADATION_RATE                1
NB_DEGRADATION_STEP             10
NB_INDIV_AGE                    20
```

```
INDIVIDUAL_EVALUATION_AGES     1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
BINDING_ZEROS_PERCENTAGE       0.75
RANDOM_BINDING_MATRIX          false
WITH_HEREDITY                  false
PROTEIN_PRESENCE_LIMIT         1e-2
ENV_SAMPLING            300
MAX_TRIANGLE_WIDTH      0.033333333
ENV_AXIS_FEATURES       METABOLISM
ENV_VARIATION           switch_in_a_list {q, or 1.5q for yoked arms}
NB_ENVIRONMENTS {2 or 4}
```

followed by the target and signal definitions of Section 3. The exploratory runs used `BACKUP_STEP` and `TREE_STEP` 10,000 as well, and an identical template otherwise. With `RANDOM_BINDING_MATRIX false` the platform reads `binding_matrix.rae` from the run directory; every run used the file distributed with `examples/dynamic_env`, unchanged. The mutation and rearrangement rates are those of `examples/complex_env`, ten times those of `examples/dynamic_env`.

3. Targets and signals

Each environment is a target function defined by three Gaussians (`ENV_ADD_GAUSSIAN` *env height mean width*):

	Gaussian 1	Gaussian 2	Gaussian 3
Target A (height, mean, width)	1.2, 0.52, 0.12	−1.4, 0.50, 0.07	0.3, 0.80, 0.03
Target B (height, mean, width)	1.2, 0.48, 0.12	−1.4, 0.50, 0.07	0.3, 0.20, 0.03

B is A reflected about 0.5. At creation the platform reports a geometric area of 0.152468 for each. Frozen arms define two copies of A (odd seeds) or of B (even seeds). The two signal proteins are taken from `examples/complex_env` (its first two `CREATE_SIGNAL` lines):

```
CREATE_SIGNAL h0 w0 h1 m1 w0 h1 m0 h0 h1 w0 h0 m1 h1 h1 m1 m0 h0 w0 h1 m1 w0 h1 m0 h0 h1 w0 h0 m1 h1 h1 m1
    m0 h1 m0 m1
CREATE_SIGNAL m0 h0 m1 h1 m1 w0 m0 m1 m0 h0 m1 h1 w0 h0 h0 h1 m1 m0 h1 w0 h1 h0 m1 h1 m0 w0 w0 m0 m1 w0 w0
    h1 h0 w0 h1 h0 h0 m0 h0 w0 h0 m1 m0 w0 h1 w0 w0 h1 m0
```

They are attached to environments with `ENV_ADD_SIGNAL` *env signal*:

Arm	Environments	`ENV_ADD_SIGNAL` lines
Informative	(A), (B)	1 1; 2 2
Yoked	(A), (A), (B), (B)	1 1; 2 2; 3 1; 4 2
No cue	(A), (B)	none

Before the first batch was launched, the platform's creation log was checked for each type of arm and reported the expected signal for each environment.

4. How environments switch

The simulations run on the platform's SIMD engine. Its environment handler (`SIMD_PhenotypicTargetHandler_R::ApplyVariation`) builds, each generation, the sequence of environments for the 20 ages of a lifetime, starting from the environment of the last age of the previous generation:

```
id_new_env = list_of_old_target_id[nb_indiv_age_ - 1];
id_old_env = list_of_old_target_id[nb_indiv_age_ - 1];
```

```
for (int16_t i = 0; i < nb_indiv_age_ ; i++) {
  double env_chang = var_prng_->random();
  if (env_chang < env_switch_probability_) {
    while (id_new_env == id_old_env)
      id_new_env = var_prng_->random(nb_env_);
    id_old_env = id_new_env;
  }
  list_env_id_[i] = id_new_env;
  ...
}
```

The environment is therefore a single Markov chain over ages that continues across generations. At each age it switches with probability q, and a switch moves to one of the *other* environments chosen uniformly by rejection sampling. The sequence is shared by the whole population. The non-SIMD handler contains a branch that, when a `deterministic` flag is set and there are two environments, gives each lifetime a fixed half-and-half split; the flag is initialised to `false` and never set, and `check_simd`, which would make the SIMD engine copy the non-SIMD sequence, is never enabled by `aevol_run`. The branch is therefore never executed.

During development, at each age the concentrations of all signal proteins are set to zero and those attached to the environment active at that age are set to 0.9, before the regulatory dynamics for that age are integrated. We checked this in the SIMD engine's network solver (`src/libaevol/7/ExpManager_7.cpp`), which is the code path used, as well as in the non-SIMD implementation (`Individual_R::EvaluateInContext`).

5. Matching switching rates in the yoked arm

Let an environment be a pair (target, signal). In the informative arm the two environments are (A, s1) and (B, s2), so every switch changes the target and the signal together. With switching probability q per age, the target changes at rate q per age and the signal at rate q per age.

In the yoked arm there are four environments, (A, s1), (A, s2), (B, s1) and (B, s2). From any of them, a switch moves to one of the three others with equal probability. Of the three, one keeps the target and changes the signal, one changes the target and keeps the signal, and one changes the two together. A switch therefore changes the target with probability 2/3 and the signal with probability 2/3. With switching probability q', each changes at rate $2q'/3$ per age, and setting $q' = 1.5q$ gives each a rate of q, as in the informative arm. Without this correction the yoked arm would fluctuate a third less often than the informative arm, and any difference between the two would be confounded with the rate of fluctuation.

The two arms also match in the marginal frequency of each signal (one half) and of each target (one half), and in the number of signal proteins present at any age (one). They differ in the association between signal and target. In the informative arm the signal determines the target, and in the yoked arm the two are independent. They also differ in the joint dynamics of a switch, since in the informative arm target and signal always change together and in the yoked arm they do so in one switch out of three. This is a direct consequence of removing the association and cannot be matched as well.

6. Measures

R-Aevol writes `stats/stats_simd_best.csv`, one row per generation, for the individual with the highest fitness. The regulatory statistics are computed in `src/libaevol/7/Stats_7.cpp`:

```
for (auto affinity: rna->affinity_list) {       // coding RNAs only
  if (affinity.enhancer_factor > 0.0) { nb_activators++; mean_activator_activity += affinity.enhancer_factor
     ; }
  if (affinity.operator_factor > 0.0) { nb_operators++;  mean_operator_activity  += affinity.operator_factor
     ; }
}
```

```
nb_influences_       = nb_operating_influences_ + nb_enhancing_influences_;
av_value_influences_ = (mean_activator_activity + mean_operator_activity) / (nb_activators + nb_operators);
```

The column `nb_link` is the number of positive enhancer factors plus the number of positive operator factors over all protein–RNA affinities (an affinity with a positive factor of each kind is counted twice), and `avg_influence` is the mean of those factors, so the total regulatory strength is $S = \texttt{nb_link} \times \texttt{avg_influence}$. The affinities include those of the signal proteins. `ExpManager_7.cpp` adds each signal to the individual's protein list and computes its affinities like those of any other protein (the condition is `prot->is_init_ || prot->signal_`), so in arms with a cue `nb_link` and S count signal-to-RNA influences, and the statistics do not separate them. When there are no links, `avg_influence` is undefined and we set $S = 0$ (an earlier, exploratory script dropped such generations instead; the difference is immaterial for the data reported, since they are rare). Genome size is `nb_functional_genes`. Each run is summarised by the mean of each measure over generations 25,001–30,000, and the run is the unit of analysis.

7. Runs and seeds

Experiment	Cells	Seeds
Exploratory (72 runs)	6 arms at $q = 0.10$, population 1024	$100000 + 1000\,a + s$, $a = 0..5$, $s = 1..12$
Pre-registered sweep, primary (168)	4 arms at $q = 0.05, 0.10, 0.20$; frozen no-cue and frozen yoked at $q = 0.10$; population 1024	$200000 + 1000\,c + s$
Pre-registered sweep, secondary (48)	fluctuating informative, yoked and no-cue, frozen informative; $q = 0.10$; population 512	$200000 + 1000\,c + s$
Grain (60)	fluctuating no-cue at $q = 0.1, 0.01, 0.001, 0.0002$; frozen no-cue	$300000 + 1000\,c + s$

The indices a and c number the arm or cell in the order written by the generator script and s the replicate. Runs were launched seed-major, so that stopping a batch early would still leave a balanced design. Every run reached 30,000 generations with exit status 0.

8. Pre-registration record

Item	Commit	Content
Sweep P1–P4	`f985ccd`	design, predictions, decision rules; `gen_params_v2.py`, `analyse_v2.py`
EvoSim confirmation P5	`4bc16c3`	viability-pilot rule, analysis script
P5 outcome	`56fe3a0`	no grid met the rule; not run
Grain G1–G4	`fa00777`	`gen_params_grain.py`, `analyse_grain.py`

Before each analysis ran, the SHA-256 checksum of the analysis script on the simulation machine was compared with the committed version (`analyse_v2.py` prefix `d0609140a970de48`; `analyse_grain.py` prefix `89a8613289731999`). Each analysis was run once, on complete data. The simulation machine's idle-sleep setting suspended runs during the first night of the sweep; this lengthened wall time but cannot change results, which are determined by the seed. From that point runs were protected by `caffeinate`. No other deviation occurred.

9. Results of every pre-registered test

Means of the two cells compared, two-sided permutation p on the difference of means (100,000 permutations), the Holm-adjusted p used for the verdicts (across the 12 tests of P1–P4, and across the four grain tests), and the p of the rank-based permutation test computed alongside. The rank-based tests lead to the same verdicts. P1 and P2 at $q = 0.05$ are listed but were not evaluable, because the manipulation check failed at that rate. Values are copied from `replication/raevol/results_v2_preregistered.txt` and `results_grain_preregistered.txt`.

Test	Cell	Mean A	Mean B	p	Holm p	Rank p
P1	$q = 0.05$	11.03	5.95	0.0046	0.0089	0.0047
P1	$q = 0.10$	13.39	7.53	0.0015	0.0046	0.0027
P1	$q = 0.20$	13.18	4.80	$< 10^{-4}$	0.0001	$< 10^{-4}$
P2	$q = 0.05$	11.03	6.32	0.0044	0.0089	0.0020
P2	$q = 0.10$	13.39	5.73	$< 10^{-4}$	0.0001	$< 10^{-4}$
P2	$q = 0.20$	13.18	4.56	$< 10^{-4}$	0.0001	$< 10^{-4}$
P3	$q = 0.05$	1.11	6.30	$< 10^{-4}$	0.0001	$< 10^{-4}$
P3	$q = 0.10$	1.25	6.30	$< 10^{-4}$	0.0001	$< 10^{-4}$
P3	$q = 0.20$	1.14	6.30	$< 10^{-4}$	0.0001	$< 10^{-4}$
P4	$q = 0.05$	26.28	16.34	$< 10^{-4}$	0.0001	$< 10^{-4}$
P4	$q = 0.10$	28.28	18.07	$< 10^{-4}$	0.0001	$< 10^{-4}$
P4	$q = 0.20$	25.60	18.02	$< 10^{-4}$	0.0003	0.0002
G1	coarse vs fine	19.24	19.45	0.801	1.000	0.885
G2	coarse vs frozen	19.24	25.58	0.0001	0.0004	0.0003
G3	coarse vs fine	2.04	2.29	0.600	1.000	0.756
G4	coarse vs frozen	2.04	8.40	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$

P1: S, fluctuating informative (A) against frozen informative (B). P2: S, fluctuating informative against fluctuating yoked. P3: S, fluctuating no-cue against frozen no-cue. P4: functional genes, frozen informative against fluctuating informative. G1 and G2: functional genes; G3 and G4: S; coarse grain ($q = 0.0002$, A) against fine grain ($q = 0.1$) or the frozen arm (B).

10. Signal-protein links (exploratory)

To separate the influences of the signal proteins from those of genome-encoded proteins, a copy of the platform was compiled with the same options after a patch to `Stats_7.cpp` (`replication/raevol/instrument_signal_links.py`). Inside the loop that counts positive enhancer and operator factors, each factor whose protein has `signal_` set is also counted and summed separately, and the two values are appended at the end of every row of `stats_simd_best.csv`. All existing columns are computed and written as before.

Resuming a run from a saved population did not reproduce the original run. A run resumed at generation 20,000 differed from the original in fitness, genome length and links from the first resumed generation on. Every run was therefore continued from its saved population at generation 30,000 to 32,000 (`decomp_launch.sh`), each in a copy of its directory, and summarised over generations 30,001–32,000 (`decomp_extract.py`). The continuations reproduced the original cell means of S (correlation 0.995 across 21 cells; ratios 0.79 to 1.16). Results are in `replication/raevol/results_decomp_exploratory.txt` and per run in `paper/data/decomp_runs.csv`.

S3 Table.

Results for every arm of every experiment

Supporting information for "Environmental fluctuation erodes gene regulation unless it can be tracked"

Values are means ± standard deviations over populations, the unit of analysis. They are generated by `paper/supplement/make_s3.py` from the per-run files `paper/data/evosim_runs.csv`, `raevol_runs.csv` and `grain_runs.csv`, which contain one row per run with the same summary measures used in the pre-registered analyses (the final 20% of each run for EvoSim and generations 25,001–30,000 for R-Aevol). R-Aevol values describe the best individual in each population, as recorded by the platform. Total regulatory strength S is the number of regulatory links times their mean strength, and is zero for generations with no links.

Table 1. EvoSim, 180 runs (grid 128, 150,000 steps, 30 seeds per arm). Populations that went extinct are counted and excluded from the means. The three frozen arms are identical runs (the cue is zero with a single state) and are listed once.

Arm	Survived	Count density	Strength density	Total strength	Nodes	Population
Fluctuating, informative	29/30	0.550 ± 0.058	0.190 ± 0.030	40.1 ± 12.2	14.38 ± 1.20	9865 ± 633
Fluctuating, yoked	29/30	0.597 ± 0.073	0.190 ± 0.026	39.3 ± 11.1	14.24 ± 1.26	9912 ± 677
Fluctuating, no cue	28/30	0.587 ± 0.065	0.184 ± 0.019	37.5 ± 11.9	14.11 ± 1.37	9899 ± 609
Frozen (all three arms)	30/30	0.467 ± 0.040	0.207 ± 0.014	38.0 ± 5.3	13.53 ± 0.78	16568 ± 2825

Time steps per generation, from the slope of mean lineage depth over the second half of each run: fluctuating arms 23.6–24.3, frozen 30.7.

Table 2. R-Aevol, exploratory experiment (72 runs; $q = 0.10$, population 1024, 30,000 generations).

Cell	n	Total strength S	Links	Genes	Metabolic error
Fluctuating, informative	12	12.68 ± 3.18	69.6 ± 19.9	19.00 ± 3.18	0.0288 ± 0.0025
Fluctuating, yoked	12	5.66 ± 1.52	39.6 ± 7.8	17.86 ± 1.91	0.0584 ± 0.0021
Fluctuating, no cue	12	1.42 ± 0.76	15.0 ± 6.6	18.85 ± 2.61	0.0626 ± 0.0005
Frozen, informative	12	4.69 ± 2.47	80.0 ± 45.3	26.12 ± 4.45	0.0078 ± 0.0014
Frozen, no cue	12	8.21 ± 5.09	68.8 ± 51.9	26.75 ± 3.96	0.0082 ± 0.0012
Frozen, yoked	12	6.17 ± 7.12	76.8 ± 63.0	24.64 ± 2.00	0.0087 ± 0.0010

Table 3. R-Aevol, pre-registered sweep (216 runs, 30,000 generations; population 1024 unless marked).

Cell	n	Total strength S	Links	Genes	Metabolic error
Fluctuating, informative, $q = 0.05$	12	11.03 ± 4.38	56.4 ± 22.9	16.34 ± 5.69	0.0382 ± 0.0361
Fluctuating, yoked, $q = 0.05$	12	6.32 ± 2.46	39.9 ± 10.6	16.36 ± 2.15	0.0547 ± 0.0032
Fluctuating, no cue, $q = 0.05$	12	1.11 ± 0.83	11.4 ± 7.2	17.45 ± 1.97	0.0601 ± 0.0012
Frozen, informative, $q = 0.05$	12	5.95 ± 3.21	94.0 ± 46.6	26.28 ± 3.48	0.0077 ± 0.0013
Fluctuating, informative, $q = 0.10$	12	13.39 ± 2.34	71.2 ± 18.6	18.07 ± 2.95	0.0293 ± 0.0035
Fluctuating, yoked, $q = 0.10$	12	5.73 ± 1.71	41.6 ± 9.5	19.94 ± 3.83	0.0583 ± 0.0021
Fluctuating, no cue, $q = 0.10$	12	1.25 ± 0.71	12.7 ± 5.1	20.03 ± 3.44	0.0627 ± 0.0007
Frozen, informative, $q = 0.10$	12	7.53 ± 4.89	91.0 ± 39.9	28.28 ± 2.69	0.0081 ± 0.0017
Fluctuating, informative, $q = 0.20$	12	13.18 ± 2.05	69.3 ± 18.0	18.02 ± 2.50	0.0303 ± 0.0030
Fluctuating, yoked, $q = 0.20$	12	4.56 ± 2.38	38.3 ± 15.0	20.75 ± 2.14	0.0626 ± 0.0016
Fluctuating, no cue, $q = 0.20$	12	1.14 ± 0.84	12.4 ± 6.4	22.43 ± 3.12	0.0641 ± 0.0004
Frozen, informative, $q = 0.20$	12	4.80 ± 2.50	83.2 ± 55.4	25.60 ± 4.03	0.0085 ± 0.0016
Frozen, no cue, $q = 0.10$	12	6.30 ± 3.09	62.4 ± 26.3	27.02 ± 3.97	0.0083 ± 0.0013
Frozen, yoked, $q = 0.10$	12	5.39 ± 1.93	86.4 ± 50.1	28.02 ± 7.27	0.0079 ± 0.0012
Fluctuating, informative, $q = 0.10$, $N = 512$	12	10.55 ± 2.50	53.3 ± 18.9	17.47 ± 3.13	0.0349 ± 0.0054
Fluctuating, yoked, $q = 0.10$, $N = 512$	12	4.29 ± 2.78	28.9 ± 12.9	17.92 ± 2.26	0.0609 ± 0.0022
Fluctuating, no cue, $q = 0.10$, $N = 512$	12	1.63 ± 0.93	16.5 ± 5.9	20.22 ± 3.53	0.0635 ± 0.0006
Frozen, informative, $q = 0.10$, $N = 512$	12	4.81 ± 2.40	59.8 ± 18.5	25.29 ± 3.08	0.0098 ± 0.0022

Table 4. R-Aevol, grain experiment (60 runs, no cue, population 1024, 30,000 generations). Mean dwell is $1/q$ ages; 20 ages make one generation.

Cell	n	Total strength S	Links	Genes	Metabolic error
Fluctuating, $q = 0.1$ (dwell 10 ages)	12	2.29 ± 1.32	17.8 ± 7.5	19.45 ± 2.22	0.0624 ± 0.0010
Fluctuating, $q = 0.01$ (dwell 100 ages)	12	1.06 ± 0.57	9.5 ± 3.1	16.25 ± 1.85	0.0594 ± 0.0018
Fluctuating, $q = 0.001$ (dwell 1,000 ages)	12	1.17 ± 0.60	13.1 ± 6.4	17.36 ± 0.95	0.0517 ± 0.0017
Fluctuating, $q = 0.0002$ (dwell 5,000 ages)	12	2.04 ± 0.98	22.9 ± 9.5	19.24 ± 1.83	0.0376 ± 0.0014
Frozen	12	8.40 ± 3.30	67.8 ± 26.6	25.58 ± 4.50	0.0081 ± 0.0015

S4 Text.

Earlier EvoSim results and the audit trail

Supporting information for "Environmental fluctuation erodes gene regulation unless it can be tracked"

This supplement records the EvoSim results that preceded the present study, what became of each, and the measurement errors found along the way together with the check that exposed each one. The complete record, including every retracted analysis, is in the project's `README.md` and `PAPER.md`.

1. Earlier EvoSim results and their status

These results came from experiments with smaller populations (mostly between 2,000 and 3,000 organisms) than the 180-run batch in the main text.

Earlier result	Evidence	Status after this study
Fluctuation increases the number of network nodes	7 of 7 frozen–fluctuating comparisons positive and significant; mean +6.3 nodes (range +2.6 to +10.2)	Reproduces published results [1,2]. R-Aevol shows the opposite (P4), at every grain tested (G1–G4), so this is a difference between the models.
Fluctuation increases connection density	Count density +0.09 to +0.14, significant in 5 of 6 comparisons; the edge increase exceeded that expected from node number alone by 27 to 41 percentage points; stable across development iterations 4 to 20	The count effect replicates (+0.12 at grid 128), but strength density falls by 11% and total strength does not change. The interpretation as denser wiring is withdrawn.
The information content of the cue does not matter	Pooled over 15 informative–scrambled comparisons, mean difference −0.17 nodes, $t = -0.30$, sign test $p = 0.61$	For genome size, reproduced in R-Aevol, where informative and yoked cues gave similar numbers of functional genes. The informative cue did change regulation in R-Aevol (P2), a measure not taken in the EvoSim regime in which organisms read the cue.
Wiring cost makes fluctuation lethal	In the sparsest regime tested, 0 of 60 frozen and 37 and 33 of 60 fluctuating populations went extinct (Fisher $p = 3.6 \times 10^{-13}$)	Consistent with the viability pilot for P5: fluctuating populations survived 2, 2 and 5 of 6 times at grids 80, 96 and 112, against 6 of 6 frozen.
Task difficulty does not set genome size	Four manipulations of the task, all null (detection floors 2.2 to 4.2 nodes)	Stands.
Organisms read the cue when development is reversible	With living organisms re-developed every few steps, switchers 0.65 against 0.23, and the best-paying channel matched in 1.5 against 1.0 of 6 states (informative against scrambled, 100 populations per arm); genome size unchanged ($p = 0.84$, 95% CI −2.1 to +2.6 nodes)	Stands. The strength measure has not been taken in this regime.

Earlier result	Evidence	Status after this study
Long-run outcome	8 runs of 25 million steps (about 630,000 generations): node number reached the minimum (12 at the time) by generation 14,000 and stayed there	Stands; the present version has a minimum of 13 because a sensor node was added.

2. Record of pre-registered predictions

Throughout the project, predictions were committed to version control before the runs that tested them.

Prediction	Outcome
Earlier EvoSim work	
Genome size rises with the complexity of the task	wrong (no change)
Rotating the yields alone makes the task conditional	wrong (inflow dominated)
An abiotic cue will be read because it cannot be depleted	wrong (legible but unread)
A conditional genome is adaptive but hard to reach by mutation	wrong (selected against)
Reversible development makes a fine-grained regime habitable	correct (survival rose from 0 of 3 to 3 of 3)
A conditional response, once evolved, raises genome size	wrong (the response evolved, size did not change)
This study	
P1–P4 (R-Aevol sweep)	all supported
P5 (EvoSim confirmation)	not run (no viable regime)
G1, G3 (grain, coarse against fine)	inconclusive
G2, G4 (grain, coarse against frozen)	refuted

3. Measurement errors and the checks that found them

Almost every wrong conclusion in the project traced back to an error of measurement, and each was found by a check whose correct answer was known in advance. They are listed because the same errors recur in simulation studies.

Error	What it produced	How it was found
Edge counts used as a measure of wiring	A density increase that is not an increase in wiring strength (this study)	Measuring summed absolute weight, prompted by the R-Aevol replication
A statistic read without checking its definition	`mean_abs_w` is summed weight divided by all n_g^2 slots, so multiplying it by the edge count gave meaningless "total strength" columns	Reading the source of the statistic before deriving from it
Parameters attributed to the wrong source	A tenfold error in the stated mutation rate of R-Aevol's configuration, in a draft of the Methods	Comparing against the platform's example files
A control that was not a control	Turning off the yield rotation also made the world poorer, and the arm with the least conditional task had the largest genomes	Measuring population size and energy in every arm
A no-op manipulation that was not a no-op	Removing a node's inputs changed its output downstream and altered every trait (repair 0.995 to 0.127)	Requiring a zero-effect arm whose result is known

Error	What it produced	How it was found
Tags that leaked	Tagging organisms by storage slot, which is reused after death, spread the tag to unrelated organisms	A control arm that reached fixation when it should not
Wrong unit of analysis	Pooling repeated measurements of the same node inflated a correlation	Reducing each node to one value
Survivorship in the sampling frame	Sampling only at the end of runs, when almost all nodes were old	Sampling during runs
A derived timescale from the wrong statistic	Generations per environmental state computed from a survivor-biased lifespan, a threefold error	Using the slope of lineage depth
Small samples	Three promising signals from 3 or 4 seeds, all gone at 12 seeds	Replication at larger n
A control checked in one mode only	The scrambled cue leaked 0.067 bits of information without a spatial gradient, the mode mostly used	Measuring its information content in every mode; fixed to 0.0004 bits
A regime that kills one arm	At grid 48, 6 of 6 fluctuating and 0 of 6 frozen populations died, which would have made the comparison one of survivors	A viability check at full run length before any batch
Identical arms counted as replicates	With a single state the cue is zero, so the three frozen arms were the same runs	Checking that arms differ
A verification script that could not fail	A scoping error kept an accumulator at its initial value, so the script would have reported success whatever the data	A negative control with a known failure
A significance test against the wrong spread	Dividing a difference by one arm's standard deviation instead of the standard error of the difference reported "no effect" for $t = 4.2$	Recomputing with Welch and permutation tests

References

1. Tsuda ME, Kawata M. Evolution of gene regulatory networks by fluctuating selection and intrinsic constraints. PLoS Comput Biol. 2010;6(8):e1000873.
2. Bentkowski P, van Oosterhout C, Mock T. A model of genome size evolution for prokaryotes in stable and fluctuating environments. Genome Biol Evol. 2015;7(8):2344–51.

S5 Table.

Prior work and how this study relates to it

Supporting information for "Environmental fluctuation erodes gene regulation unless it can be tracked"

The table sets out which parts of the argument are established in the literature, which conflict with our results, and which we did not find precedent for. Literature searches covered theory of plasticity and cue reliability, gene-regulatory-network evolution under fluctuating selection, digital evolution (Avida, Aevol), network sparseness and modularity, and weighted network measures.

Claim	Status	Sources and relation to this study
Reliable cues favour plastic responses; without them, fluctuation favours generalists, bet-hedging or genetic tracking	Established	Levins 1968 [1]; Moran 1992 [2]; Kussell and Leibler 2005 [3]; Botero et al. 2015 [4]; Ashander et al. 2016 [23], in which responding to a poorly predictive cue amplifies mismatch. Our results concern the corresponding effect on the regulatory network.
In slowly changing environments populations track the environment genetically instead of evolving plasticity	Established	Botero et al. 2015 [4]; Rago et al. 2019 [21]. Consistent with the better match to the target, without a change in regulation, at coarse grain in R-Aevol.
In a gene regulatory network model, plasticity evolves mainly under fast, erratic change with reliable cues	Established	Jorritsma and van den Berg 2026 [5]. Their abstract reports which parts of the network become sensitive to mutation. We have not been able to read the full text; if it reports connection strength by cue reliability, part of our architectural result would overlap with it.
In a network with a dedicated sensor gene, discrete plasticity evolves as the sensor comes to activate some genes and repress others	Published	Sakamoto and Innan 2024 [19]. The sensor is always informative and gene number does not evolve, so neither an uninformative-cue control nor genome size is examined. EvoSim's sensor node follows the same design.
Fluctuating environments can select for or against gene regulation	Established experimentally	Pollack, Nozoe and Kussell 2025 [6], who report that catabolite symport flux strongly affects selection for regulation and note that laboratory evolution under metabolic fluctuation often loses the regulation of metabolic genes.
Fluctuation without information about the optimum promotes complex regulatory networks	Published; conflicts with our R-Aevol results for a random-walk optimum	Tsuda and Kawata 2010 [7]. Their optimum moved across generations, and the networks received an on/off signal that carried no information about its position. The effect was strong for a random walk and slight, limited to large and infrequent shifts, for an optimum alternating between two positions, which is the pattern of our design. Our grain experiment found that grain does not reconcile the two results. The remaining candidate explanations are the novel optima of the random walk and differences in the genotype–phenotype map.
Fluctuating environments favour larger genomes	Published; matches earlier EvoSim results	Bentkowski et al. 2015 [8]. R-Aevol shows the opposite.

Claim	Status	Sources and relation to this study
Genome size is limited by selection for mutational robustness and falls with population size and mutation rate	Established in Aevol	Knibbe et al. 2007 [9]; Luiselli et al. 2024 [10]. Consistent with the small genomes in large EvoSim populations.
Sensing and non-sensing populations evolve differently in fluctuating environments	Published, different measures	Lalejini et al. 2021 [11] (Avida), measuring trait loss and the accumulation of deleterious mutations. Network architecture was not measured.
Network models with and without a cue input compared	Published	Ng and Kinjo 2023 [22] and 2024 [20]; plasticity-led evolution appears only with the cue input, and network density is fixed by the mutation operator. Removing the cue changes its presence and its information together, the confound the yoked control removes.
Regulatory complexity is measured by counting genes or edges	Common practice	Tsuda and Kawata 2010 [7] (gene numbers, degree distributions); Espinosa-Soto 2018 [12] (binary edges, weights not discussed).
Network size and connectivity can grow without selection for them	Established	Soyer and Bonhoeffer 2006 [13].
The number of transcription factors scales faster than genome size and is higher in free-living bacteria	Established; two interpretations	Comparative genomics links it to lifestyle and environmental variety (van Nimwegen 2003 [16]; Cases, de Lorenzo and Ouzounis 2003 [17]). R-Aevol simulations reproduce the scaling and attribute it to mutation and rearrangement rates instead of lifestyle (Beslon et al. 2010 [18]).
Summed edge strength as a network measure	Standard method	Iwasaki, Tsuda and Kawata 2013 [14].
The yoked control	Established in the study of learning	Church 1964 [15] describes a bias that arises when individual subjects are paired; our arms are not paired, so it does not apply.
Not found in the literature		
Edge count does not track edge strength under fluctuation	This study	EvoSim (exploratory; more edges with the same total strength), R-Aevol informative-cue populations (pre-registered secondary analysis; no more links but stronger regulation).
A yoked control that separates a cue's information from its presence in the evolution of regulatory networks	This study	R-Aevol, with the rates of target and signal change matched between arms (S2 Text). Varying cue reliability achieves the same separation if it includes a cue that is present but uninformative; the abstract of [5] does not say whether it does.
Regulatory strength and genome size compared across informative, yoked and absent cues	This study	Pre-registered predictions P1–P4; subject to the caveat on [5] above.

References

1. Levins R. Evolution in Changing Environments. Princeton: Princeton University Press; 1968.
2. Moran NA. The evolutionary maintenance of alternative phenotypes. Am Nat. 1992;139(5):971–89.
3. Kussell E, Leibler S. Phenotypic diversity, population growth, and information in fluctuating environments. Science. 2005;309(5743):2075–8.

4. Botero CA, Weissing FJ, Wright J, Rubenstein DR. Evolutionary tipping points in the capacity to adapt to environmental change. Proc Natl Acad Sci USA. 2015;112(1):184–9.
5. Jorritsma A, van den Berg P. The evolution of plasticity and evolvability in a simple gene regulatory network. Evolution. 2026;80(1):230–9.
6. Pollack D, Nozoe T, Kussell E. Gene regulation, phenotypic memory, and selection in fluctuating environments. bioRxiv [Preprint]. 2025. doi:10.1101/2025.06.23.661119.
7. Tsuda ME, Kawata M. Evolution of gene regulatory networks by fluctuating selection and intrinsic constraints. PLoS Comput Biol. 2010;6(8):e1000873.
8. Bentkowski P, van Oosterhout C, Mock T. A model of genome size evolution for prokaryotes in stable and fluctuating environments. Genome Biol Evol. 2015;7(8):2344–51.
9. Knibbe C, Coulon A, Mazet O, Fayard JM, Beslon G. A long-term evolutionary pressure on the amount of noncoding DNA. Mol Biol Evol. 2007;24(10):2344–53.
10. Luiselli J, Rouzaud-Cornabas J, Lartillot N, Beslon G. Genome streamlining: effect of mutation rate and population size on genome size reduction. Genome Biol Evol. 2024;16(12):evae250.
11. Lalejini A, Ferguson AJ, Grant NA, Ofria C. Adaptive phenotypic plasticity stabilizes evolution in fluctuating environments. Front Ecol Evol. 2021;9:715381.
12. Espinosa-Soto C. On the role of sparseness in the evolution of modularity in gene regulatory networks. PLoS Comput Biol. 2018;14(5):e1006172.
13. Soyer OS, Bonhoeffer S. Evolution of complexity in signaling pathways. Proc Natl Acad Sci USA. 2006;103(44):16337–42.
14. Iwasaki WM, Tsuda ME, Kawata M. Genetic and environmental factors affecting cryptic variations in gene regulatory networks. BMC Evol Biol. 2013;13:91.
15. Church RM. Systematic effect of random error in the yoked control design. Psychol Bull. 1964;62:122–31.
16. van Nimwegen E. Scaling laws in the functional content of genomes. Trends Genet. 2003;19(9):479–84.
17. Cases I, de Lorenzo V, Ouzounis CA. Transcription regulation and environmental adaptation in bacteria. Trends Microbiol. 2003;11(6):248–53.
18. Beslon G, Parsons DP, Sanchez-Dehesa Y, Peña JM, Knibbe C. Scaling laws in bacterial genomes: a side-effect of selection of mutational robustness? BioSystems. 2010;102(1):32–40.
19. Sakamoto T, Innan H. On the evolutionary origin of discrete phenotypic plasticity. G3 (Bethesda). 2024;14(9):jkae144.
20. Ng ETH, Kinjo AR. Plasticity-led and mutation-led evolutions are different modes of the same developmental gene regulatory network. PeerJ. 2024;12:e17102.
21. Rago A, Kouvaris K, Uller T, Watson R. How adaptive plasticity evolves when selected against. PLoS Comput Biol. 2019;15(3):e1006260.
22. Ng ETH, Kinjo AR. Plasticity-led evolution as an intrinsic property of developmental gene regulatory networks. Sci Rep. 2023;13(1):19830.
23. Ashander J, Chevin LM, Baskett ML. Predicting evolutionary rescue via evolving plasticity in stochastic environments. Proc R Soc B. 2016;283(1839):20161690.

S6 Text.

EvoSim: connection count and wiring strength under fluctuation (exploratory)

Supporting information for "Environmental fluctuation erodes gene regulation unless it can be tracked"

This study began with EvoSim, an individual-based model in which the genome is a regulatory network of evolvable size (S1 Text). Its earlier results are summarised in S4 Text. This experiment measured the strength of regulatory connections as well as their number. It is exploratory. The pre-registered confirmation could not be run (section 4), and EvoSim organisms did not use the cue in this regime, so the experiment concerns fluctuation that cannot be tracked. The main text reports the R-Aevol experiments.

1. Model

EvoSim is an individual-based model of unicellular organisms on a two-dimensional lattice with diffusing, cross-fed resources (S1 Text). Each genome is a gene regulatory network, specified by a weight matrix and node biases, with 13 to 32 nodes. Thirteen nodes are fixed. Twelve are read out as phenotypic traits (resource-uptake allocation, body size, motility, division threshold, DNA repair, stress tolerance and leakage) and one receives an abiotic cue, much like the sensor gene of Sakamoto and Innan [1]. The remaining hidden nodes are added by duplication and removed by deletion, so genome size can evolve. The network is developed once at birth, and the phenotype does not change afterwards. Mutation can create and remove connections and each non-zero connection has a small maintenance cost, so connectivity is also free to evolve.

The design follows the R-Aevol experiments with an abiotic cue. Fluctuating arms have an informative cue, a yoked cue decoupled from the environmental state by running on an incommensurate period, or no cue. In a frozen environment the cue is zero, so the three frozen arms are identical.

EvoSim is described in full, following the ODD protocol [2], in S1 Text; the parts relevant to this experiment are summarised below. Organisms live on a $G \times G$ lattice ($G = 128$) with six diffusing resource channels. External inflow, which varies seasonally, enters the first channel (with the exception described under Environment and cue), and consuming channel r releases a byproduct into channel $r + 1$, with yields declining along this chain. All arms also experience rare random mass-mortality and famine events, on average one every 25,000 time steps. Each organism has an energy store and divides when the store exceeds a heritable threshold. Starvation and the other causes of death are described in S1 Text.

Genome and development. A genome is a weight matrix W and a bias vector b over n_g nodes, with $13 \le n_g \le 32$. Nodes 1 to 12 are read out as the phenotype, node 13 receives the abiotic cue, and hidden nodes $14, \ldots, n_g$ have no fixed meaning. Development takes place once, at birth. Expression is initialised at 0.1 and updated eight times as

$$e_k \leftarrow \tanh\Big(b_k + \sum_m W_{mk}\, e_m + u_k\Big), \tag{1}$$

where u_k is the local concentration of resource k for $k \le 6$, the cue for $k = 13$, and zero otherwise. The final expression values are mapped to the twelve traits.

Mutation. At each birth, a Poisson number of point mutations with mean $10^{-3} n_g^2$ is applied, each adding $\mathcal{N}(0, 0.25^2)$ to a randomly chosen weight. This rate is reduced by up to 85% by the organism's heritable repair trait. With probability equal to the reduced per-weight rate times n_g, one node bias receives a mutation of the same size. The structural mutations are gene silencing (probability 0.002), which sets a node's outgoing weights to zero; gene duplication (0.002), which copies the regulatory role of one node onto another and adds $\mathcal{N}(0, 0.05^2)$ noise to every copied weight; and the addition or deletion of a hidden node (0.003 each), where an added node is a duplicate of an existing one made in the same way. Connection silencing (probability 0.2 per birth) sets one randomly chosen weight to exactly zero. Without

it, additive mutation would drive density towards one. Each hidden node costs 0.0015 and each non-zero weight 5×10^{-5} energy units per time step.

Environment and cue. An abiotic condition cycles through discrete states with a period of 2,000 time steps. Each state changes which resource channel gives the highest yield, and in fluctuating arms 40% of the external inflow goes to the favoured channel. In frozen arms all inflow enters channel 1, which then also gives the highest yield. The phase of the cycle, scaled to $[-1, 1]$, is the input to node 13 and identifies the current state. Fluctuating arms have six states and frozen arms one. In the yoked arm the cue is instead the phase multiplied by $10/\varphi \approx 6.18$, with φ the golden ratio, and wrapped back into the same range, so that it takes the same values but carries only 0.0004 bits of information about the state. In the no-cue arm the gain of the cue is zero. With a single state the cue is identically zero, so the three frozen arms produce identical runs and were analysed as one arm.

Measures. For every organism at birth we record the number of weights with $|w| > 10^{-4}$, the strength density $\sum |w|/n_g^2$ over all n_g^2 weights, including zeros, and n_g. The log holds their population means at each time step. Count density is the mean number of non-zero weights divided by the squared mean n_g, and total strength is the mean strength density multiplied by the squared mean n_g. Each measure is then averaged over the last 20% of the run. Populations that went extinct were excluded and their number is reported.

Runs. We ran six arms with 30 seeds each for 150,000 time steps (180 runs), about 6,200 generations in fluctuating and 4,900 in frozen populations (24 and 31 time steps per generation, from the slope of mean lineage depth). Runs are deterministic for a given seed and, on the same machine and Julia version, bit-identical across thread counts (checked at 1, 4 and 8 threads).

2. Results

Earlier EvoSim experiments had found that fluctuation increases connection density (S4 Text). We repeated the comparison in 180 runs (six arms of 30 seeds, 150,000 time steps each) and measured the strength of connections as well as their number. The cue is zero in a frozen environment, so the three frozen arms are identical and the runs contain 120 distinct populations. Without a cue, fluctuation raised count density, the proportion of possible connections that are non-zero, from 0.467 to 0.587 (+25.5%, $p < 10^{-4}$, $d = 2.2$; 28 surviving fluctuating and 30 frozen populations; Fig AA), and the number of non-zero connections from 86 to 120 (+39%, $p < 10^{-4}$). Total wiring strength did not change (38.0 in frozen and 37.5 in fluctuating populations, $p = 0.86$; Fig AC). Fluctuating networks thus had more connections but not more wiring, and the average non-zero connection was about 30% weaker. Strength density, the summed absolute weight divided by the number of possible connections, fell from 0.207 to 0.184 (−10.9%, $p < 10^{-4}$, $d = -1.4$; Fig AB), mostly because there were more possible connections. Node count stayed close to its minimum of 13 in this regime and increased only slightly (+0.6 nodes, $p = 0.047$), which raises n_g^2 by about 9%. With the informative and the yoked cue, fluctuation had effects in the same directions (count density +18% and +28%, total strength unchanged, $p = 0.40$ and 0.59).

Population size is a possible confound. Fluctuating populations were 40% smaller than frozen ones, as expected since in frozen dishes all inflow feeds the best-paying channel (S1 Text), and population size can change genome architecture by itself. Because the population sizes of the two arms did not overlap, the two effects cannot be separated statistically, and the correlations within arms point in different directions for the two measures. Larger populations had lower strength density in every arm (r between −0.31 and −0.47; Fig AD), so population size would predict higher strength density in the smaller fluctuating populations, the reverse of what we found. Count density, in contrast, was higher in smaller populations within each fluctuating arm (r between −0.63 and −0.72), though not in the frozen arm ($r = 0.24$), so population size may contribute to the increase in connection number. The cue was also not used in this

regime. Populations with the informative and the yoked cue were the same size ($p = 0.79$) and had the same strength density ($p = 0.99$), although count density was lower with the informative cue (0.55 against 0.60, $p = 0.009$). EvoSim organisms respond to the cue only when development is reversible, and even then they match their phenotype to the current state only in part (S4 Text). The EvoSim results concern fluctuation that the organisms cannot track.

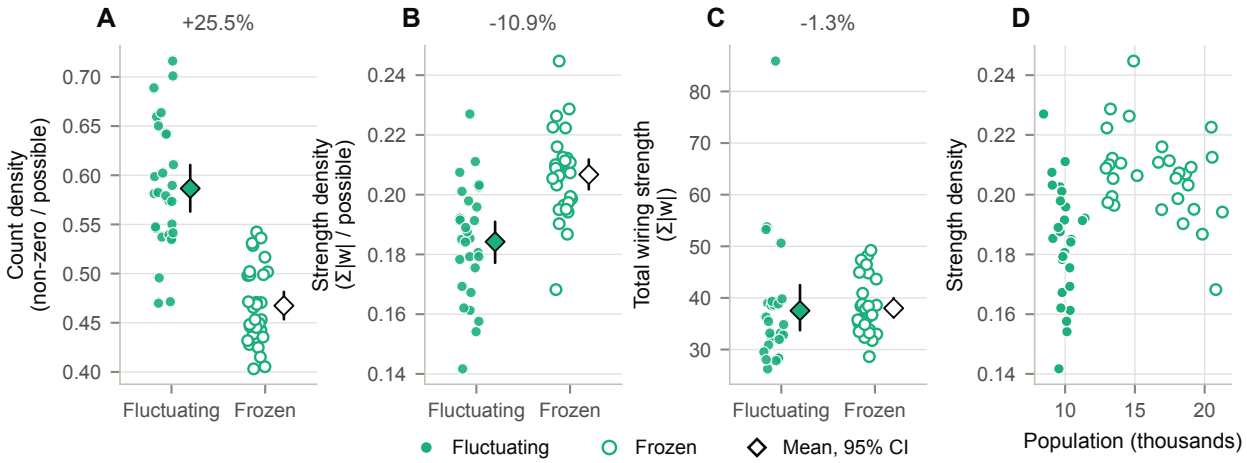


Fig A. EvoSim: more connections, no more wiring. Fluctuating (no cue; filled) versus frozen (hollow) populations, one point per population; diamonds are means with bootstrap 95% confidence intervals. Numbers above panels are the change in the mean. (A) Count density, the number of non-zero connections divided by the number of possible connections. (B) Strength density, the summed absolute weight divided by the number of possible connections. (C) Total wiring strength. (D) Strength density against population size within each arm. Extinct populations (2 of 30 fluctuating, 0 of 30 frozen) are excluded.

3. Interpretation

Gene duplication and node addition copy the connections of an existing node and add small random values to every copied weight, including weights that were zero, so each duplication can create many weak connections. Point mutations that hit a zero weight also create connections. If purifying selection is weaker in the smaller populations of fluctuating environments, such connections would persist and connection number would rise without an increase in total strength, which fits the correlations with population size within the fluctuating arms. We have not tested this mechanism.

4. Limitations and the confirmation that was not run

Fluctuation and population size are confounded between arms. The within-arm correlations argue against population size as the cause of the lower strength density but are consistent with a contribution to the higher connection count. We pre-registered a confirmation of the divergence between count and strength (P5; commit `4bc16c3`) in a regime with more hidden nodes, on condition that fluctuating and frozen populations survived a viability pilot there. They did not (2 to 5 of 6 fluctuating populations survived at grids 80, 96 and 112, compared with 6 of 6 frozen populations; commit `56fe3a0`), so the confirmation was not run and the result remains exploratory.

References

1. Sakamoto T, Innan H. On the evolutionary origin of discrete phenotypic plasticity. G3 (Bethesda). 2024;14(9):jkae144.
2. Grimm V, Railsback SF, Vincenot CE, Berger U, Gallagher C, DeAngelis DL, et al. The ODD protocol for describing agent-based and other simulation models: a second update to improve clarity, replication, and structural realism. J Artif Soc Soc Simul. 2020;23(2):7.